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Element H: Prototype Testing and Data Collection Plan

Testing Details (H1, H2, H3, H4)

	Element C High Priority requirement	testing setup and procedure	test cases and number of trials	expected range of results	pass/fail criteria, goal
ex	The prototype must remain intact when held upside down.	1) Hold the prototype in a hand using the grip in this photo. <insert grip photo> 2) Take 3 seconds to slowly rotate the prototype around the indicated axis into an upside down position with a smooth motion. <insert photo with annotation of axes of rotation> 3) Note any observations about prototype integrity. 4) Take 3 seconds to rotate the prototype in reverse, back to its original position,	a. Rotate around x-axis (10 trials) b. Rotate around y-axis (10 trials) c. Rotate around z-axis (10 trials)	Parts of the prototype may: – remain unchanged – deform – shift position – tear – detach	Fail: one or more parts of the prototype tears or detaches Pass: zero parts of the prototype tear or detach Goal: all parts of the prototype remain unchanged when the



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		using a smooth motion			prototype is returned to its original position
ex	mass of prototype < 3kg	1) Tare a digital scale. 2) Place the prototype securely on the scale. 3) Wait for the reading on the scale to stabilize before recording the value.	test cases: N/A # trials: 10	0kg<reading<10 kg	Fail: mean of the trials ≥ 3 kg Pass: mean of the trials < 3kg Goal: mean of the trials < 2kg
1	The program must evaluate each device and recommend delaying or turning off, or keeping the device on.	Input 10 devices into the program with a mix of essential and non-essential classifications. Run the program once during a peak time (e.g., 'evening') and once during an off-peak time (e.g., 'morning').	Test cases: n/a Trials: 7	Essential devices should always return "OK to stay ON". Non-essential devices should return "Recommend delaying" during peak hours and "OK to stay ON" during off-peak hours.	Fail: one output is incorrect. Pass: all 5 devices are correctly categorized based on essentiality and peak status. Goal: Ensure the program applies energy-saving logic accurately and reliably for each condition.



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2	The program must allow user input to define each device's name, power usage (in kWh), and whether it is essential or non-essential.	Input at least 10 devices with different names, power usage values, and essentiality. Verify input is correctly stored and used in decision-making.	Test cases: n/a Trials: 4	All inputs must be retained without error. Power usage should be numeric. Device names should appear correctly in output messages.	Pass: the program stores all inputs and reflects them accurately in the output. Fail: there's a mismatch between input and output. Goal: validate that the system captures and utilizes data.
3	The program must categorize time of day as either "peak" or "off-peak" hours based on user input and adjust recommendati	Run the program multiple times using different time inputs: "morning", "afternoon", "evening", and "night". Track how the program classifies each time.	Test cases: n/a Trials: 3	Expected Range of Results: "afternoon" and "evening" = peak "morning" and "night" = off-peak	Pass: time categorization is consistent with defined rules and recommendations change with time. Fail: time input is misclassified



Name:Anthony

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	ons accordingly.				or ignored. Goal: Ensure that time-based energy logic functions properly and consistently guides recommendatio n output.
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Name: Anthony Claudet**Date:** _____**Class:** _____

Feedback From Field Experts (H5)

Name of your field expert:

Catherine C

Why are they a field expert?

She studied physics at Imperial College London and works as a software engineer and part time game developer. She is very experienced with coding and the many concepts related to my program, such as data storage, logical computing, and systems. She codes in C# and has worked with complex software such as Unity.

What did they say to you about the tests you are proposing?

Automatically categorizing different times of the day as peak or off-peak is a very realistic approach. For future reference, integrating real utility rate schedules or dynamic pricing APIs would make it even better. I'd also recommend eventually testing edge cases, like devices that toggle between essential and non-essential use, as energy priorities can change based on context.

Revised Testing Details That Incorporate Field Expert Feedback (H5)

Will consider adding more complexities to devices, such as the ability to transition between an essential and non-essential status. I will also attempt to integrate different factors that may affect energy pricing at different times.

